

Centrality in Soccer Passing Networks

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Data source

As a data source, I would use StatsBomb Open Data, which is free event-level data from professional soccer matches published on GitHub. It includes recent tournament data. Every pass is recorded with the passer and the recipient, so each team's game becomes a passing network:

- **Nodes:** players.
- **Edges:** connect two players from the same team, weighted by the number of passes between them.

For categorical variables, every player has a position, which can, for example, be grouped into defender, midfielder, and forward/attacker. Each match has a result (win, draw, or loss) and stats like dribbles, shots, and expected goals (xG), which can be used as outcomes.

StatsBomb Open Data: <https://github.com/hudl/open-data>

Plan for loading the data

1. Install the statsbombpy package (or just read the raw JSON files from GitHub) and list the available competitions.
2. Pull the match list for Copa America 2024, then load the event data for each match.
3. Filter events by completed passes and keep the passer, the recipient, and the team.
4. Build one weighted directed graph per team and per match in networkx, with players as nodes and pass counts as edge weights.
5. Add each player's position group as a node attribute, using the lineup data.
6. Calculate degree centrality for every player in every match, then compare position groups with ANOVA or Kruskal-Wallis.

Hypothetical outcome

My hypothesis is that midfielders will have the highest degree centrality, defenders second, and forwards likely the lowest. Midfielders are kind of the glue between the defensive line and the attacking line, so most passing sequences run through them.

- **Positive outcome:** players with higher degree centrality should be involved in more attacking sequences, which could predict more shots and higher expected goals (xG) for their team.

- **Negative outcome:** if a team's defenders have higher centrality than its midfielders, the team may be stuck passing around the back instead of moving up the pitch. Teams who deeply rely on ball possession. Also more defensive teams that tend to have a more direct style of playing, which a lot of times excludes the midfielders from the building phase (direct passes between defenders and attackers). That pattern could predict fewer chances created and a higher chance of losing.

It is important to normalize by minutes on the pitch, since a player's centrality depends on minutes played. Another approach could be to include only players who played the whole game. However, that would exclude key players who might (and actually are) subbed off with only a couple of minutes left to play to include an extra defender, a more defensive midfielder, or even to replace an attacker with a midfielder.